

Packet & Log Analysis

1 **ssh_lab.pcap** → Network capture

Type: Packet capture (from **tcpdump**)

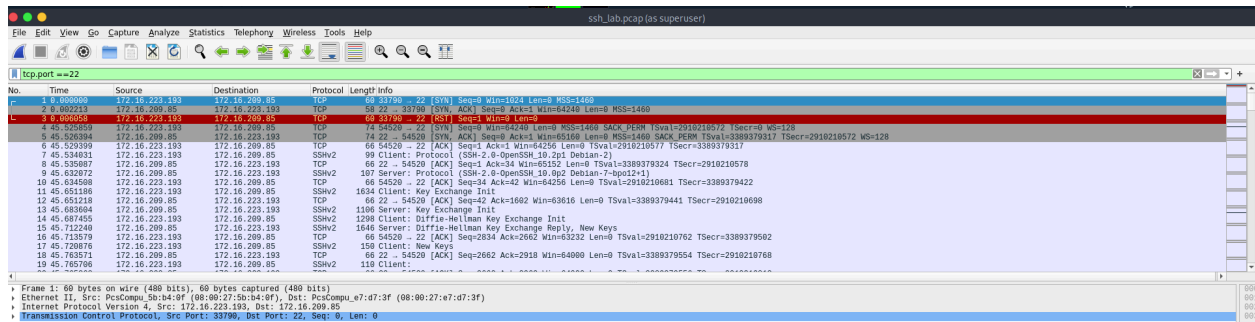
Purpose: Contains SSH or TCP session data (e.g., between Kali ↔ Parrot)

- ◆ **Analyze with Wireshark (GUI)**

sudo wireshark /tmp/artifacts/ssh_lab.pcap &

Then apply filters like:

- **tcp.port == 22** → show only SSH traffic



- **ip.addr == 172.16.223.193 && ip.addr == 172.16.209.85** → show both systems

No.	Time	Source	Destination	Protocol	Length	Info
2	0.000000	172.16.223.193	172.16.209.85	TCP	60	33790 → 22 [SYN] Seq=33790 Win=0 Len=0 MSS=1460
3	0.006958	172.16.223.193	172.16.209.85	TCP	60	33790 → 22 [RST] Seq=33790 Win=0 Len=0
4	4.529399	172.16.223.193	172.16.209.85	TCP	74	54520 → 22 [SYN] Seq=0 Win=64248 Len=0 MSS=1460 SACK_PERM TSval=2910210572 TSecr=0 WS=128
5	4.529894	172.16.223.193	172.16.209.85	TCP	74	22 → 54520 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3389379317 TSecr=2910210572 WS=128
6	4.529399	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2910210577 TSecr=3389379317
7	4.534031	172.16.223.193	172.16.209.85	SSHv2	89	Client: Protocol (SSH-2.0-OpenSSH_5.2p1 Debian-2)
8	4.535087	172.16.223.193	172.16.209.85	TCP	66	22 → 54520 [ACK] Seq=1 Ack=34 Win=85152 Len=0 TSval=3389379324 TSecr=2910210578
9	4.535072	172.16.209.85	172.16.223.193	SSHv2	167	Server: Protocol (SSH-2.0-OpenSSH_5.8p2 Debian-7ubuntu2-1)
10	4.534508	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=34 Ack=42 Win=64256 Len=0 TSval=2910210681 TSecr=3389379422
11	4.551186	172.16.223.193	172.16.209.85	SSHv2	1634	Client: Key Exchange Init
12	4.551218	172.16.209.85	172.16.223.193	TCP	66	22 → 54520 [ACK] Seq=42 Ack=1602 Win=63616 Len=0 TSval=3389379441 TSecr=2910210698
13	4.563604	172.16.209.85	172.16.223.193	SSHv2	1106	Server: Key Exchange Init
14	4.567455	172.16.223.193	172.16.209.85	SSHv2	1298	Client: Diffie-Hellman Key Exchange Init
15	4.572249	172.16.209.85	172.16.223.193	SSHv2	1646	Server: Diffie-Hellman Key Exchange Reply, New Keys
16	4.573578	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=594 Ack=2662 Win=63332 Len=0 TSval=2910210762 TSecr=3389379502
17	4.572876	172.16.223.193	172.16.209.85	SSHv2	150	Client: New Keys
18	4.763571	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=2662 Ack=2918 Win=64000 Len=0 TSval=3389379554 TSecr=2910210768
19	4.765706	172.16.223.193	172.16.209.85	SSHv2	110	Client:

- `tcp.flags.syn == 1` → show connection attempts (handshakes)

No.	Time	Source	Destination	Protocol	Length	Info
2	0.000000	172.16.223.193	172.16.209.85	TCP	60	33790 → 22 [SYN] Seq=33790 Win=0 Len=0 MSS=1460
3	0.006958	172.16.223.193	172.16.209.85	TCP	60	33790 → 22 [RST] Seq=33790 Win=0 Len=0
4	4.529399	172.16.223.193	172.16.209.85	TCP	74	54520 → 22 [SYN] Seq=0 Win=64248 Len=0 MSS=1460 SACK_PERM TSval=2910210572 TSecr=0 WS=128
5	4.529894	172.16.209.85	172.16.223.193	TCP	74	22 → 54520 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3389379317 TSecr=2910210572 WS=128
32	49.432776	172.16.223.193	172.16.209.85	TCP	74	54524 → 22 [SYN] Seq=0 Win=64248 Len=0 MSS=1460 SACK_PERM TSval=2910214481 TSecr=0 WS=128
33	49.432943	172.16.209.85	172.16.223.193	TCP	74	22 → 54524 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3389390823 TSecr=2910214401 WS=128
58	53.255449	172.16.223.193	172.16.209.85	TCP	74	32854 → 22 [SYN] Seq=0 Win=64248 Len=0 MSS=1460 SACK_PERM TSval=2910218298 TSecr=0 WS=128
59	53.255422	172.16.209.85	172.16.223.193	TCP	74	22 → 32854 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3389397846 TSecr=2910218298 WS=128
86	57.892142	172.16.223.193	172.16.209.85	TCP	74	32866 → 22 [SYN] Seq=0 Win=64248 Len=0 MSS=1460 SACK_PERM TSval=2910222101 TSecr=0 WS=128
87	57.892384	172.16.209.85	172.16.223.193	TCP	74	22 → 32866 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3389390844 TSecr=2910222101 WS=128
112	60.348555	172.16.223.193	172.16.209.85	TCP	74	57689 → 22 [SYN] Seq=0 Win=64248 Len=0 MSS=1460 SACK_PERM TSval=2910225889 TSecr=0 WS=128
113	60.348605	172.16.209.85	172.16.223.193	TCP	74	22 → 57689 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3389394631 TSecr=2910225889 WS=128
138	64.617166	172.16.223.193	172.16.209.85	TCP	74	57616 → 22 [SYN] Seq=0 Win=64248 Len=0 MSS=1460 SACK_PERM TSval=2910229606 TSecr=0 WS=128
139	64.617893	172.16.209.85	172.16.223.193	TCP	74	22 → 57616 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3389396468 TSecr=2910229666 WS=128
164	68.387542	172.16.223.193	172.16.209.85	TCP	74	57626 → 22 [SYN] Seq=0 Win=64248 Len=0 MSS=1460 SACK_PERM TSval=2910233436 TSecr=0 WS=128
165	68.387628	172.16.209.85	172.16.223.193	TCP	74	22 → 57626 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3389402178 TSecr=2910233436 WS=128
190	72.156806	172.16.223.193	172.16.209.85	TCP	74	59914 → 22 [SYN] Seq=0 Win=64248 Len=0 MSS=1460 SACK_PERM TSval=2910237199 TSecr=0 WS=128
191	72.156864	172.16.209.85	172.16.223.193	TCP	74	22 → 59914 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3389405841 TSecr=2910237199 WS=128
217	111.683178	172.16.223.193	172.16.209.85	TCP	74	42668 → 22 [SYN] Seq=0 Win=64248 Len=0 MSS=1460 SACK_PERM TSval=2910276733 TSecr=0 WS=128

2 transfer_only.pcap

Type: Likely filtered capture (e.g., file transfer or smaller subset)

Purpose: Focused network activity capture.

No.	Time	Source	Destination	Protocol	Length	Info
2	0.000000	172.16.223.193	172.16.209.85	TCP	60	33790 → 22 [SYN] Seq=33790 Win=0 Len=0 MSS=1460
3	0.006958	172.16.223.193	172.16.209.85	TCP	60	33790 → 22 [RST] Seq=33790 Win=0 Len=0
4	4.529399	172.16.223.193	172.16.209.85	TCP	74	54520 → 22 [SYN] Seq=0 Win=64248 Len=0 MSS=1460 SACK_PERM TSval=2910210572 TSecr=0 WS=128
5	4.529894	172.16.209.85	172.16.223.193	TCP	74	22 → 54520 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3389379317 TSecr=2910210572 WS=128
6	4.534031	172.16.223.193	172.16.209.85	SSHv2	89	Client: Protocol (SSH-2.0-OpenSSH_5.2p1 Debian-2)
7	4.534508	172.16.223.193	172.16.209.85	SSHv2	1634	Client: Key Exchange Init
8	4.535087	172.16.223.193	172.16.209.85	SSHv2	1298	Client: Diffie-Hellman Key Exchange Init
9	4.535072	172.16.209.85	172.16.223.193	SSHv2	1646	Server: Diffie-Hellman Key Exchange Reply, New Keys
10	4.572876	172.16.223.193	172.16.209.85	SSHv2	150	Client: New Keys, Encrypted packet (len=68)
11	4.763571	172.16.223.193	172.16.209.85	SSHv2	110	Client: Encrypted packet (len=44)
12	4.765706	172.16.223.193	172.16.209.85	SSHv2	134	Client: Encrypted packet (len=88)
13	4.817649	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=3030 Ack=2970 Win=63488 Len=0 TSval=2910210866 TSecr=3389379564
14	4.836331	172.16.223.193	172.16.209.85	SSHv2	214	Client: Encrypted packet (len=148)
15	4.840569	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=3178 Ack=3022 Win=63488 Len=0 TSval=2910214469 TSecr=3389383210
16	4.841111	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=3178 Ack=3022 Win=63488 Len=0 TSval=2910214471 TSecr=3389383210
17	4.842094	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=3178 Ack=3023 Win=63488 Len=0 TSval=2910214474 TSecr=3389383215
18	4.842776	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=3178 Ack=3023 Win=63488 Len=0 TSval=2910214480 TSecr=3389383219
19	4.851583	172.16.223.193	172.16.209.85	TCP	66	54524 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2910214488 TSecr=3389383223

transfer_only.pcap (as superuser)

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ip.dst == 172.16.209.85

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	172.16.223.193	172.16.209.85	TCP	60	33790 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2	0.000058	172.16.223.193	172.16.209.85	TCP	60	33790 → 22 [RST] Seq=1 Win=0 Len=0
3	45.525859	172.16.223.193	172.16.209.85	TCP	74	54520 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460
4	45.529399	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TS
5	45.534031	172.16.223.193	172.16.209.85	SSHv2	99	Client: Protocol (SSH-2.0-OpenSSH_10.2p1 Debian
6	45.634508	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=34 Ack=42 Win=64256 Len=0
7	45.651186	172.16.223.193	172.16.209.85	SSHv2	1634	Client: Key Exchange Init
8	45.687455	172.16.223.193	172.16.209.85	SSHv2	1298	Client: Diffie-Hellman Key Exchange Init
9	45.713579	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=2834 Ack=2662 Win=63232 Le
10	45.720876	172.16.223.193	172.16.209.85	SSHv2	150	Client: New Keys, Encrypted packet (len=68)
11	45.765706	172.16.223.193	172.16.209.85	SSHv2	110	Client: Encrypted packet (len=44)
12	45.768518	172.16.223.193	172.16.209.85	SSHv2	134	Client: Encrypted packet (len=68)
13	45.817849	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=3030 Ack=2970 Win=63488 Le
14	45.833631	172.16.223.193	172.16.209.85	SSHv2	214	Client: Encrypted packet (len=148)
15	49.420569	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=3178 Ack=3022 Win=63488 Le
16	49.421711	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [FIN, ACK] Seq=3178 Ack=3022 Win=634
17	49.426094	172.16.223.193	172.16.209.85	TCP	66	54520 → 22 [ACK] Seq=3179 Ack=3023 Win=63488 Le
18	49.432776	172.16.223.193	172.16.209.85	TCP	74	54524 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460
19	49.451503	172.16.223.193	172.16.209.85	TCP	66	54524 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TS

Frame 5: 99 bytes on wire (792 bits), 99 bytes captured (792 bits) on interface unknown, id 0
 Ethernet II, Src: PcsCompu_5b:b4:0f (08:00:27:5b:b4:0f), Dst: PcsCompu_e7:d7:3f (08:00:27:e7:d7:3f)
 Internet Protocol Version 4, Src: 172.16.223.193, Dst: 172.16.209.85
 Transmission Control Protocol, Src Port: 54520, Dst Port: 22, Seq: 1, Ack: 1, Len: 33
 SSH Protocol

3 auth_full.log

Type: SSH authentication log (from `journalctl`)

Purpose: Shows SSH login attempts (accepted/failed), user logins, timestamps, and IPs.

◆ View and search

`less /tmp/artifacts/auth_full.log`

```
Nov 07 23:54:19 parrot systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Nov 07 23:54:19 parrot sshd[3180]: Server listening on 0.0.0.0 port 22.
Nov 07 23:54:19 parrot sshd[3180]: Server listening on :: port 22.
Nov 07 23:54:19 parrot systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Nov 08 00:03:44 parrot sshd-session[3690]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=172.16.223.193 user=testuser
Nov 08 00:03:44 parrot sshd-session[3690]: pam_unix(sshd:auth): getting password (testuser)
Nov 08 00:03:44 parrot sshd-session[3690]: pam_unix(sshd:auth): request whologonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)!
Nov 08 00:03:44 parrot sshd-session[3690]: pam_unix(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'testuser')
Nov 08 00:03:44 parrot sshd-session[3690]: Failed password for testuser from 172.16.223.193 port 59674 ssh2
Nov 08 00:03:49 parrot sshd-session[3690]: Connection closed by authenticating user testuser 172.16.223.193 port 59674 (preauth)
Nov 08 00:03:49 parrot sshd-session[3692]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=172.16.223.193 user=testuser
Nov 08 00:03:49 parrot sshd-session[3692]: pam_unix(sshd:auth): getting password (testuser)
Nov 08 00:03:49 parrot sshd-session[3692]: pam_unix(sshd:auth): request whologonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)!
Nov 08 00:03:49 parrot sshd-session[3692]: pam_unix(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'testuser')
Nov 08 00:03:51 parrot sshd-session[3692]: Failed password for testuser from 172.16.223.193 port 45596 ssh2
Nov 08 00:03:53 parrot sshd-session[3692]: Connection closed by authenticating user testuser 172.16.223.193 port 45596 (preauth)
Nov 08 00:03:54 parrot sshd-session[3694]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=172.16.223.193 user=testuser
Nov 08 00:03:54 parrot sshd-session[3694]: pam_unix(sshd:auth): getting password (testuser)
Nov 08 00:03:54 parrot sshd-session[3694]: pam_unix(sshd:auth): request whologonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)!
Nov 08 00:03:54 parrot sshd-session[3694]: pam_unix(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'testuser')
Nov 08 00:03:55 parrot sshd-session[3694]: Failed password for testuser from 172.16.223.193 port 45604 ssh2
Nov 08 00:03:56 parrot sshd-session[3694]: Connection closed by authenticating user testuser 172.16.223.193 port 45604 (preauth)
Nov 08 00:03:56 parrot sshd-session[3696]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=172.16.223.193 user=testuser
Nov 08 00:03:56 parrot sshd-session[3696]: pam_unix(sshd:auth): getting password (testuser)
Nov 08 00:03:56 parrot sshd-session[3696]: pam_unix(sshd:auth): request whologonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)!
Nov 08 00:03:56 parrot sshd-session[3696]: pam_unix(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'testuser')
Nov 08 00:03:59 parrot sshd-session[3696]: Failed password for testuser from 172.16.223.193 port 60424 ssh2
```

◆ Extract failed logins

`grep "Failed password" /tmp/artifacts/auth_full.log`

```
Applications Places System Parrot Terminal
File Edit View Search Terminal Help
[jadhav@parrot]~$
[jadhav@parrot]~$ grep "Failed password" /tmp/artifacts/auth_full.log
Nov 08 00:03:46 parrot sshd-session[3690]: Failed password for testuser from 172.16.223.193 port 59674 ssh2
Nov 08 00:03:51 parrot sshd-session[3692]: Failed password for testuser from 172.16.223.193 port 45596 ssh2
Nov 08 00:03:55 parrot sshd-session[3694]: Failed password for testuser from 172.16.223.193 port 45604 ssh2
Nov 08 00:03:59 parrot sshd-session[3696]: Failed password for testuser from 172.16.223.193 port 60424 ssh2
Nov 08 00:04:03 parrot sshd-session[3698]: Failed password for testuser from 172.16.223.193 port 60432 ssh2
Nov 08 00:04:05 parrot sshd-session[3708]: Failed password for testuser from 172.16.223.193 port 60446 ssh2
Nov 08 00:04:07 parrot sshd-session[3710]: Failed password for testuser from 172.16.223.193 port 60462 ssh2
Nov 10 12:33:01 parrot sshd-session[2857]: Failed password for testuser from 172.16.223.193 port 54658 ssh2
Nov 10 12:33:01 parrot sshd-session[2858]: Failed password for testuser from 172.16.223.193 port 54674 ssh2
Nov 10 12:33:06 parrot sshd-session[2858]: Failed password for testuser from 172.16.223.193 port 54674 ssh2
Nov 11 18:53:04 parrot sshd-session[3452]: Failed password for user from 172.16.223.193 port 48300 ssh2
Nov 11 18:53:04 parrot sshd-session[3451]: Failed password for user from 172.16.223.193 port 48294 ssh2
Nov 11 18:53:07 parrot sshd-session[3452]: Failed password for user from 172.16.223.193 port 48300 ssh2
Nov 11 18:53:10 parrot sshd-session[3452]: Failed password for user from 172.16.223.193 port 48300 ssh2
Nov 11 18:53:10 parrot sshd-session[3451]: Failed password for user from 172.16.223.193 port 48294 ssh2
Nov 11 18:53:13 parrot sshd-session[3452]: Failed password for user from 172.16.223.193 port 48300 ssh2
Nov 11 18:53:13 parrot sshd-session[3451]: Failed password for user from 172.16.223.193 port 48294 ssh2
Nov 11 18:53:16 parrot sshd-session[3452]: Failed password for user from 172.16.223.193 port 48300 ssh2
Nov 11 18:53:16 parrot sshd-session[3451]: Failed password for user from 172.16.223.193 port 48294 ssh2
Nov 11 18:53:19 parrot sshd-session[3451]: Failed password for user from 172.16.223.193 port 48294 ssh2
Nov 11 18:53:19 parrot sshd-session[3452]: Failed password for user from 172.16.223.193 port 48300 ssh2
Nov 11 18:53:22 parrot sshd-session[3457]: Failed password for user from 172.16.223.193 port 55198 ssh2
Nov 11 18:53:22 parrot sshd-session[3458]: Failed password for user from 172.16.223.193 port 55212 ssh2
Nov 11 18:53:25 parrot sshd-session[3457]: Failed password for user from 172.16.223.193 port 55198 ssh2
Nov 11 18:53:25 parrot sshd-session[3458]: Failed password for user from 172.16.223.193 port 55212 ssh2
Nov 11 18:53:28 parrot sshd-session[3457]: Failed password for user from 172.16.223.193 port 55198 ssh2
Nov 11 18:53:28 parrot sshd-session[3458]: Failed password for user from 172.16.223.193 port 55212 ssh2
Nov 11 18:53:31 parrot sshd-session[3457]: Failed password for user from 172.16.223.193 port 55198 ssh2
Nov 11 18:53:31 parrot sshd-session[3458]: Failed password for user from 172.16.223.193 port 55212 ssh2
```

Extract successful logins

grep "Accepted password" /tmp/artifacts/auth_full.log

```
File Edit View Search Terminal Help
[jadhav@parrot]~$
[jadhav@parrot]~$ grep "Accepted password" /tmp/artifacts/auth_full.log
Nov 08 00:04:28 parrot sshd-session[3712]: Accepted password for testuser from 172.16.223.193 port 44602 ssh2
Nov 08 00:05:27 parrot sshd-session[3844]: Accepted password for testuser from 172.16.223.193 port 44960 ssh2
Nov 08 00:05:45 parrot sshd-session[3935]: Accepted password for testuser from 172.16.223.193 port 60748 ssh2
Nov 10 12:33:03 parrot sshd-session[2857]: Accepted password for testuser from 172.16.223.193 port 54658 ssh2
Nov 10 12:33:06 parrot sshd-session[2952]: Accepted password for testuser from 172.16.223.193 port 40892 ssh2
Nov 10 12:34:37 parrot sshd-session[3853]: Accepted password for testuser from 172.16.223.193 port 38790 ssh2
Nov 11 19:35:23 parrot sshd-session[4833]: Accepted password for user from 172.16.223.193 port 53404 ssh2
Nov 11 22:12:15 parrot sshd-session[6409]: Accepted password for user1 from 172.16.223.193 port 46515 ssh2
Nov 11 22:36:59 parrot sshd-session[6966]: Accepted password for user1 from 172.16.223.193 port 47642 ssh2
Nov 12 08:25:52 parrot sshd-session[5472]: Accepted password for testuser from 172.16.223.193 port 35304 ssh2
Nov 12 08:27:06 parrot sshd-session[5614]: Accepted password for testuser from 172.16.223.193 port 43014 ssh2
Nov 12 08:27:55 parrot sshd-session[5704]: Accepted password for testuser from 172.16.223.193 port 39132 ssh2
[jadhav@parrot]~$
```

◆ Count failed login attempts

grep -c "Failed password" /tmp/artifacts/auth_full.log

```
[jadhav@parrot]~$
[jadhav@parrot]~$ grep -c "Failed password" /tmp/artifacts/auth_full.log
2222
[jadhav@parrot]~$
```

4 auth.log

Type: Traditional syslog-style auth log (if created)

Purpose: Same as above but in Ubuntu-style format.

grep "sshd" /tmp/artifacts/auth.log | less

```
File Edit View Search Terminal Help
Nov 12 08:10:12 parrot sshd[1045]: Received signal 15; terminating.
Nov 12 08:10:13 parrot sshd[5121]: Server listening on 0.0.0.0 port 22.
Nov 12 08:10:13 parrot sshd[5121]: Server listening on :: port 22.
Nov 12 08:23:31 parrot sshd-session[5423]: Connection from 172.16.223.193 port 54520 on 172.16.209.85 port 22 rdomain ""
Nov 12 08:23:31 parrot sshd-session[5423]: pam_unix(sshd:auth): authentication failure; logname=uid=0 euid=0 tty=ssh ruser= rhost=172.16.223.193 user=testuser
Nov 12 08:23:31 parrot sshd-session[5423]: pam_winbind(sshd:auth): getting password (0x00000388)
Nov 12 08:23:31 parrot sshd-session[5423]: pam_winbind(sshd:auth): pam_get_item returned a password
Nov 12 08:23:31 parrot sshd-session[5423]: pam_winbind(sshd:auth): request wbcLogonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)!
Nov 12 08:23:31 parrot sshd-session[5423]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'testuser')
Nov 12 08:23:33 parrot sshd-session[5423]: Failed password for testuser from 172.16.223.193 port 54520 ssh2
Nov 12 08:23:34 parrot sshd-session[5423]: Connection closed by authenticating user testuser 172.16.223.193 port 54520 [preauth]
Nov 12 08:23:34 parrot sshd[5121]: srclimit_penalise: ipv4: new 172.16.223.193/32 deferred penalty of 5 seconds for penalty: failed authentication
Nov 12 08:23:34 parrot sshd-session[5426]: Connection from 172.16.223.193 port 54524 on 172.16.209.85 port 22 rdomain ""
Nov 12 08:23:35 parrot sshd-session[5426]: pam_unix(sshd:auth): authentication failure; logname=uid=0 euid=0 tty=ssh ruser= rhost=172.16.223.193 user=testuser
Nov 12 08:23:35 parrot sshd-session[5426]: pam_winbind(sshd:auth): getting password (0x00000388)
Nov 12 08:23:35 parrot sshd-session[5426]: pam_winbind(sshd:auth): pam_get_item returned a password
Nov 12 08:23:35 parrot sshd-session[5426]: pam_winbind(sshd:auth): request wbcLogonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)!
Nov 12 08:23:35 parrot sshd-session[5426]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'testuser')
Nov 12 08:23:37 parrot sshd-session[5426]: Failed password for testuser from 172.16.223.193 port 54524 ssh2
Nov 12 08:23:38 parrot sshd-session[5426]: Connection closed by authenticating user testuser 172.16.223.193 port 54524 [preauth]
Nov 12 08:23:38 parrot sshd-session[5428]: Connection from 172.16.223.193 port 32854 on 172.16.209.85 port 22 rdomain ""
Nov 12 08:23:38 parrot sshd-session[5428]: pam_unix(sshd:auth): authentication failure; logname=uid=0 euid=0 tty=ssh ruser= rhost=172.16.223.193 user=testuser
Nov 12 08:23:38 parrot sshd-session[5428]: pam_winbind(sshd:auth): getting password (0x00000388)
Nov 12 08:23:38 parrot sshd-session[5428]: pam_winbind(sshd:auth): pam_get_item returned a password
Nov 12 08:23:38 parrot sshd-session[5428]: pam_winbind(sshd:auth): request wbcLogonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)!
Nov 12 08:23:38 parrot sshd-session[5428]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'testuser')
Nov 12 08:23:40 parrot sshd-session[5428]: Failed password for testuser from 172.16.223.193 port 32854 ssh2
Nov 12 08:23:42 parrot sshd-session[5428]: Connection closed by authenticating user testuser 172.16.223.193 port 32854 [preauth]
Nov 12 08:23:42 parrot sshd-session[5430]: Connection from 172.16.223.193 port 32866 on 172.16.209.85 port 22 rdomain ""
Nov 12 08:23:42 parrot sshd-session[5430]: pam_unix(sshd:auth): authentication failure; logname=uid=0 euid=0 tty=ssh ruser= rhost=172.16.223.193 user=testuser
Nov 12 08:23:42 parrot sshd-session[5430]: pam_winbind(sshd:auth): getting password (0x00000388)
```

Compare with:

diff /tmp/artifacts/auth.log /tmp/artifacts/auth_full.log

```

File Edit View Search Terminal Help
Nov 11 22:42:32 parrot sshd-session[7156]: pam_winbind(sshd:auth): request wbcLogonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)!
Nov 11 22:42:32 parrot sshd-session[7156]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'user1')
Nov 11 22:42:34 parrot sshd-session[7156]: Failed password for user1 from 172.16.223.193 port 46584 ssh2
Nov 11 22:42:35 parrot sshd-session[7156]: pam_winbind(sshd:auth): getting password (0x00000388)
Nov 11 22:42:35 parrot sshd-session[7156]: pam_winbind(sshd:auth): pam_get_item returned a password
Nov 11 22:42:35 parrot sshd-session[7156]: pam_winbind(sshd:auth): request wbcLogonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)!
Nov 11 22:42:35 parrot sshd-session[7156]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'user1')
Nov 11 22:42:38 parrot sshd-session[7156]: Failed password for user1 from 172.16.223.193 port 46584 ssh2
Nov 11 22:42:39 parrot sshd-session[7156]: pam_winbind(sshd:auth): getting password (0x00000388)
Nov 11 22:42:39 parrot sshd-session[7156]: pam_winbind(sshd:auth): pam_get_item returned a password
Nov 11 22:42:39 parrot sshd-session[7156]: pam_winbind(sshd:auth): request wbcLogonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)!
Nov 11 22:42:39 parrot sshd-session[7156]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'user1')
Nov 11 22:42:41 parrot sshd-session[7156]: Failed password for user1 from 172.16.223.193 port 46584 ssh2
Nov 11 22:42:42 parrot sshd-session[7156]: error: maximum authentication attempts exceeded for user1 from 172.16.223.193 port 46584 ssh2 [preauth]
Nov 11 22:42:42 parrot sshd-session[7156]: Disconnecting authenticating user user1 172.16.223.193 port 46584: Too many authentication failures [preauth]
Nov 11 22:42:42 parrot sshd-session[7156]: PAM 5 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=172.16.223.193 user=user1
Nov 11 22:42:42 parrot sshd-session[7156]: PAM service(sshd) ignoring max retries; 6 > 3
Nov 11 22:42:42 parrot sshd-session[7160]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=172.16.223.193 user=user1
Nov 11 22:42:42 parrot sshd-session[7160]: pam_winbind(sshd:auth): getting password (0x00000388)
Nov 11 22:42:42 parrot sshd-session[7160]: pam_winbind(sshd:auth): pam_get_item returned a password
-- Boot 4c7c31142cd942c38425f0c2e84a1f08 --
Nov 11 23:38:40 parrot systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Nov 11 23:38:41 parrot sshd[1039]: Server listening on 0.0.0.0 port 22.
Nov 11 23:38:41 parrot sshd[1039]: Server listening on :: port 22.
Nov 11 23:38:41 parrot systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
-- Boot 4c7c31142cd942c38425f0c2e84a1f08 --
Nov 12 07:16:38 parrot systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Nov 12 07:16:38 parrot sshd[1045]: Server listening on 0.0.0.0 port 22.
Nov 12 07:16:38 parrot sshd[1045]: Server listening on :: port 22.
Nov 12 07:16:38 parrot systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
--(x)~[jadhav@parrot]~[~]

```

5 last.log

Type: Output of `last -n 20`

Purpose: Shows **successful logins** (username, IP, time, duration).

`cat /tmp/artifacts/last.log`

```
Parrot Terminal
File Edit View Search Terminal Help

--[jadhav@parrot]--[~]rot sshd-session[7156]: pam_winbind(sshd:auth): request
-- $cat /tmp/artifacts/last.log PAM error: PAM_AUTHINFO_UNAVAIL (9)!
jadhav11 tty7:39 parrot:0 sshd-session[7Wed Nov 12 07:16](:still logged innal
reboot PAsystem bootUN6.12.32-amd64r = Wed Nov 12 07:16 still running
jadhav11 tty7:41 parrot:0 sshd-session[7Tue Nov 11 23:39](- crash (07:37)rom 1
reboot4 system boot 6.12.32-amd64 Tue Nov 11 23:38 still running
user1 11 pts/2:42 parrot172.16.223.193[7Tue Nov 11 22:36](- crash (01:01)on at
jadhav11 tty7 172.16.2:0.193 port 46584Tue Nov 11 17:52 - crash (05:45)
reboot11 system bootn6.12.32-amd64on[7Tue Nov 11 17:48]ngstill runningg use
jadhavporttty7584: Too :0ny authenticatiTue Nov 11 14:29](- crash (03:19)
reboot11 system bootn6.12.32-amd64on[7Tue Nov 11 14:28]atstill runningfailur
jadhav=0 tty7ssh ruser:0rhost=172.16.22Tue Nov 11 13:54](- crash (00:34)
reboot11 system bootn6.12.32-amd64on[7Tue Nov 11 13:54](:still running max r
jadhav11 tty7:42 parrot:0 sshd-session[7Mon Nov 10 12:13](- crash (1+01:40)icat
rebootin-system booty=6.12.32-amd64ost=Mon Nov 10 12:12 usestill running
jadhav11 22:42:42 parrot sshd-session[7Fri Nov 7 23:53](- crash (2+12:19)ing
jadhav tty7 :0 Fri Nov 7 23:46 - crash (2+12:26)
reboot11 system bootn6.12.32-amd64on[7Fri Nov 7 23:45](:still runningget_
jadhav tty7 :0 Fri Nov 7 20:18 - crash (03:26)
rebootboot system boot386.12.32-amd645aa7Fri Nov 7 20:18 still running
jadhav11 tty78:40 parrot:0 systemd[1]: StFri Nov 7 20:17](- crash (00:01)re Sh
reboot11 system bootn6.12.32-amd64 SeFri Nov 7 20:16](:still running2.
Nov 11 23:38:41 parrot sshd[1039]: Server listening on *: port 22.
tmp begins Mon4Sepa8o13:02:54-2025 Started ssh.service - OpenBSD Secure She
--[jadhav@parrot]--[~]942c38425f0c2e84a1f08 --
-- $ 12 07:16:38 parrot systemd[1]: Starting ssh.service - OpenBSD Secure Sh
Nov 12 07:16:38 parrot sshd[1045]: Server listening on 0.0.0.0 port 22.
```

Look for:

- pts/0 sessions
- remote IPs
- logout times

6,lastb.log

Type: Output of `lastb -n 50`

Purpose: Shows **failed login attempts**.

Useful for brute-force detection.

`cat /tmp/artifacts/lastb.log`

```
File Edit View Search Terminal Help
[jadhav@parrot]~$ cat /tmp/artifacts/lastb.log
jadhav tty7 0 Wed Nov 12 07:16 still logged in
testuser ssh:notty 172.16.223.193 Wed Nov 12 08:24 - 08:24 (00:00)
testuser ssh:notty 172.16.223.193 Wed Nov 12 08:24 - 08:24 (00:00)
testuser ssh:notty 172.16.223.193 Wed Nov 12 08:24 - 08:24 (00:00)
testuser ssh:notty 172.16.223.193 Wed Nov 12 08:23 - 08:23 (00:00)
testuser ssh:notty 172.16.223.193 Wed Nov 12 08:23 - 08:23 (00:00)
testuser ssh:notty 172.16.223.193 Wed Nov 12 08:23 - 08:23 (00:00)
testuser ssh:notty 172.16.223.193 Wed Nov 12 08:23 - 08:23 (00:00)
testuser ssh:notty 172.16.223.193 Wed Nov 12 08:23 - 08:23 (00:00)
testuser ssh:notty 172.16.223.193 Wed Nov 12 08:23 - 08:23 (00:00)
testuser ssh:notty 172.16.223.193 Wed Nov 12 08:23 - 08:23 (00:00)
testuser ssh:notty 172.16.223.193 Wed Nov 12 08:23 - 08:23 (00:00)
user1 ssh:notty 172.16.223.193 Tue Nov 11 22:43 - 22:43 (00:00)
user1 ssh:notty 172.16.223.193 Tue Nov 11 22:43 - 22:43 (00:00)
user1 ssh:notty 172.16.223.193 Tue Nov 11 22:43 - 22:43 (00:00)
user1 ssh:notty 172.16.223.193 Tue Nov 11 22:42 - 22:42 (00:00)
user1 ssh:notty 172.16.223.193 Tue Nov 11 22:42 - 22:42 (00:00)
user1 ssh:notty 172.16.223.193 Tue Nov 11 22:42 - 22:42 (00:00)
user1 ssh:notty 172.16.223.193 Tue Nov 11 22:42 - 22:42 (00:00)
user1 ssh:notty 172.16.223.193 Tue Nov 11 22:42 - 22:42 (00:00)
user1 ssh:notty 172.16.223.193 Tue Nov 11 22:42 - 22:42 (00:00)
user1 ssh:notty 172.16.223.193 Tue Nov 11 22:42 - 22:42 (00:00)
user1 ssh:notty 172.16.223.193 Tue Nov 11 22:42 - 22:42 (00:00)
user1 ssh:notty 172.16.223.193 Tue Nov 11 22:42 - 22:42 (00:00)
user1 ssh:notty 172.16.223.193 Tue Nov 11 22:42 - 22:42 (00:00)
user1 ssh:notty 172.16.223.193 Tue Nov 11 22:42 - 22:42 (00:00)
```

7,secret_stat.txt

Type: Output of `stat` and `ls -l` on `/home/testuser/secret.txt`

Purpose: File metadata — owner, permissions, modification times.

`cat /tmp/artifacts/secret_stat.txt`


```

[jadhav@parrot]~$ cat /tmp/artifacts/secret_stat.txt
File: /home/testuser/secret.txt
Size: 27 bytes  Blocks: 8/93  IO Block: 4096  regular file
Device: 0,39  Inode: 116719  Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1003/testuser)  Gid: ( 1003/testuser)
Access: 2025-11-07 23:58:07.018875910 +0530  Nov 12 08:23 - 08:23  (00:00)
Modify: 2025-11-12 08:13:46.334472311 +0530  Nov 12 08:23 - 08:23  (00:00)
Change: 2025-11-12 08:13:55.538458346 +0530  Nov 12 08:23 - 08:23  (00:00)
 Birth: 2025-11-07 23:58:07.018875910 +0530  Nov 12 08:23 - 08:23  (00:00)
-rw-r--r-- 1 testuser testuser 27 2025-11-12 08:13:46.334472311 +0530 /home/testuser/secret.txt
testuser ssh:notty 172.16.223.193 Wed Nov 12 08:23 - 08:23 (00:00)
[jadhav@parrot]~$ cat /tmp/artifacts/secret_stat.txt
testuser ssh:notty 172.16.223.193 Wed Nov 12 08:23 - 08:23 (00:00)

```

8, tcpdump.pid

Type: PID file for tcpdump process

Purpose: Contains the process ID of the capture that was running.

cat /tmp/artifacts/tcpdump.pid

```

[jadhav@parrot]~$ cat /tmp/artifacts/tcpdump.pid
2541  ssh:notty 172.16.223.193 Tue
[jadhav@parrot]~$ cat /tmp/artifacts/tcpdump.pid
2541  ssh:notty 172.16.223.193 Tue

```

9,auth_filtered_iso.log

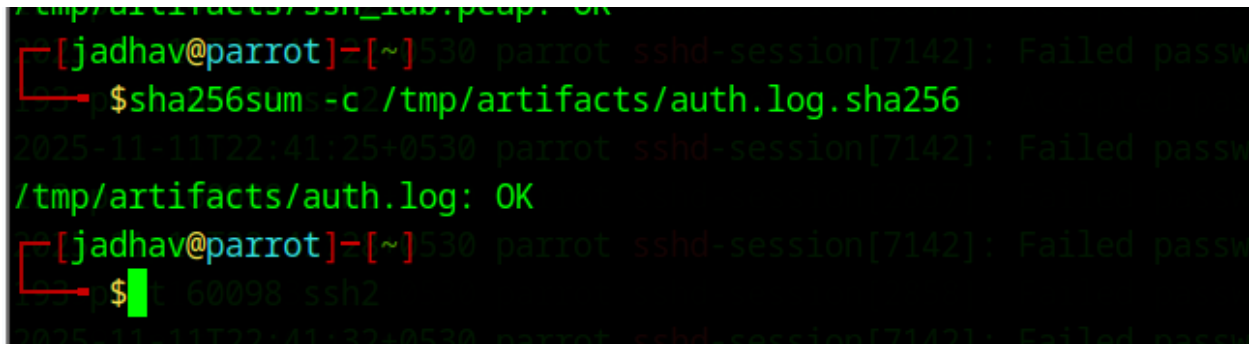
Type: Filtered version of auth log (possibly containing only SSH-related events)

Purpose: Focused analysis.

grep "sshd" /tmp/artifacts/auth_filtered_iso.log

10. ssh_lab.pcap.sha256 and auth.log.sha256

sha256sum -c /tmp/artifacts/auth.log.sha256



Summary Table

File	Type	Command
ssh_lab.pcap	Network capture	wireshark /tmp/artifacts/ssh_lab.pcap
transfer_only.pcap	Partial capture	tshark -r /tmp/artifacts/transfer_only.pcap
auth_full.log	SSH logs	grep "Failed password" /tmp/artifacts/auth_full.log
auth.log	System auth log	grep "sshd" /tmp/artifacts/auth.log
last.log	Successful logins	cat /tmp/artifacts/last.log
lastb.log	Failed logins	cat /tmp/artifacts/lastb.log
ss_22.txt	SSH sockets	cat /tmp/artifacts/ss_22.txt
lsof_ssh.txt	SSH processes	cat /tmp/artifacts/lsof_ssh.txt
secret_stat.txt	File metadata	cat /tmp/artifacts/secret_stat.txt
tcpdump.pid	Capture PID	cat /tmp/artifacts/tcpdump.pid
auth_filtered_iso.log	Filtered log	grep sshd /tmp/artifacts/auth_filtered_iso.log
bcs0121.c	C source file	cat /tmp/artifacts/bcs0121.c
*.sha256	Hashes	sha256sum -c file.sha256